SaaS Customer Churn Analysis & Prediction Pipeline

End-to-end business case study using Python, SQL, Power BI and ML

5,000	14.4%	$429.2K	$649K
Customers	**Churn Rate**	**Total MRR**	**ARR at Risk**
Cleaned records	720 churned	Monthly recurring revenue	From actual churned customers

PORTFOLIO REPORT

This report presents the full project workflow as a recruiter-ready and executive-readable case study: raw data ingestion, Python data preparation, machine learning churn scoring, SQL business analysis, Power BI dashboarding, and retention recommendations.

Prepared by: Khushnuma Ansari

Suggested GitHub folder: reports/SaaS_Churn_Prediction_Business_Case_Report.pdf

Executive Summary

The project moves SaaS churn analysis from backward-looking reporting to proactive customer-retention intelligence. It uses Python to clean and engineer features, machine learning to estimate churn risk, SQL to create decision-ready business questions, and Power BI to communicate the results through a four-page dashboard.

5,000
Customers analyzed
Final Power BI dataset

720
Actual churned customers
Observed target variable

577
High/Critical accounts
Predicted high-risk portfolio

$520K
High/Critical ARR
Potential retention focus

BUSINESS MEANING

- Monthly contracts are the riskiest contract group, with churn nearly double the annual contract churn rate.
- Free plan customers show the highest churn rate, while Enterprise customers are more stable but carry higher revenue value.
- Competitor movement and missing product capabilities are the most common recorded churn reasons.
- The dashboard converts model scores into risk lists and business segments that Customer Success teams can act on.

Business Problem

SaaS companies depend on recurring revenue. When customers churn, the business loses not only monthly revenue but also future expansion potential. The goal of this project is to identify churn patterns early and create a workflow that helps business users prioritize retention actions.

Project Objectives

- Build an end-to-end analytics pipeline from raw SaaS customer data to a Power BI reporting dataset.
- Clean, transform, and enrich raw customer records in Python.
- Train and compare churn prediction models to score customers by churn probability.
- Use SQL to answer business questions related to churn rate, revenue risk, contract type, plan, industry, and customer health.
- Create a Power BI dashboard that leadership and Customer Success teams can use for monitoring and action.

Business Questions Answered

- Which plan and contract segments churn most often?
- Which churn reasons explain the largest number of lost customers?
- Which industries contribute the most annual revenue at risk?
- Which customer health and engagement signals are most predictive of churn?
- Which active accounts should be prioritized for retention outreach?

Dataset Audit and Cleaning

Stage	Rows	Columns	Purpose
Raw dataset	5,030	34	Original SaaS churn source file
Cleaned dataset	5,000	46	Prepared for modeling and SQL
Power BI dataset	5,000	51	Final dataset with predictions

Key Cleaning Actions

- Removed 30 duplicate customer records to keep one row per customer.
- Converted date fields and tenure fields into analysis-ready data types.
- Handled missing numeric values such as NPS, CSAT, session duration, ticket resolution time, and tenure.
- Standardized country labels such as US/United States, UK/United Kingdom, and case variations.
- Created a clean CSV that became the common input for machine learning, SQL, and Power BI.

Raw Missing Value Snapshot

Column	Missing Values
tenure_months	20
avg_session_duration_min	100
avg_ticket_resolution_hrs	121
nps_score	200
csat_score	151

Note: churn_date and churn_reason are naturally blank for active customers, so they are not treated as data quality failures.

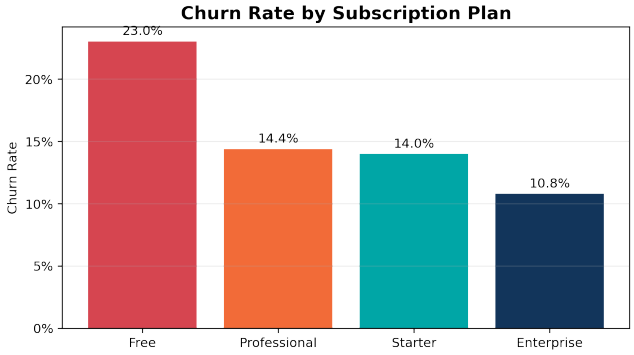
Feature Engineering

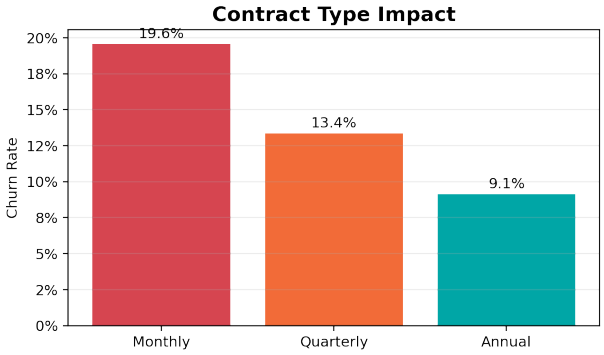
Feature engineering turned raw customer records into business-readable indicators and model-ready variables. These features help translate technical fields into customer health, revenue risk, engagement quality, and retention priority.

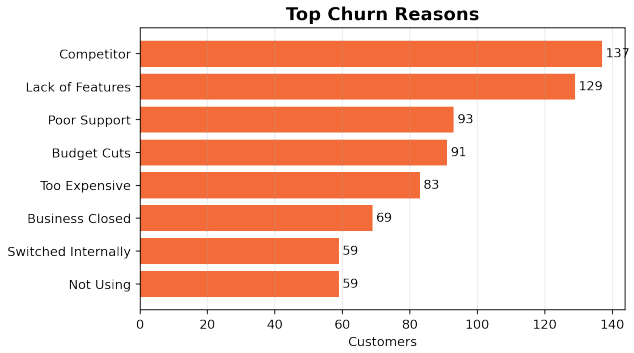
Feature	Business Purpose
revenue_at_risk	Revenue exposure from churned customers
customer_lifetime_days	Customer age from signup to churn/reference date
login_frequency	Activity segment from 90-day login count
risk_sagment	Business risk classification such as Stable or At Risk
nps_category	Detractor, Passive, or Promoter customer satisfaction band

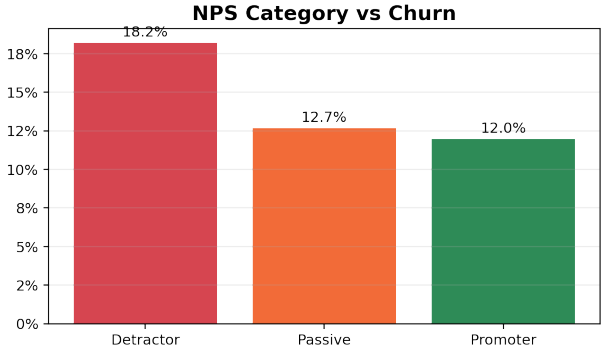
Feature	Business Purpose
support_burden	Tickets per tenure month, indicating service friction
revenue_per_member	Account value normalized by active team size
is_high_value	Flag for accounts with higher commercial value
churn_probability	Predicted churn probability from the ML layer
predicted_risk_level	Low, Medium, High, or Critical action band

EDA Highlights: Where Churn Concentrates





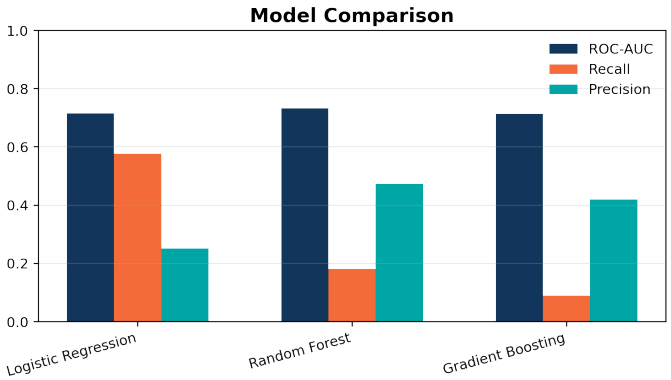


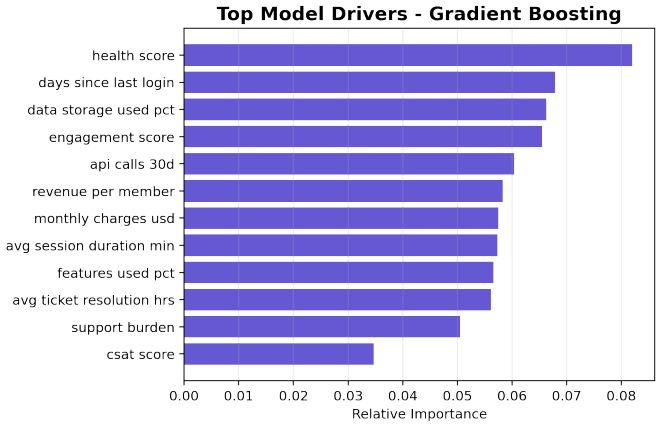


- Free plan churn rate is 23.0%, the highest plan-level churn observed.
- Monthly contract churn is 19.6%, compared with 9.1% for annual contracts.
- Detractors churn at 18.2%, confirming customer sentiment as a meaningful retention signal.

Machine Learning Layer

The modeling notebook prepared a supervised churn prediction workflow using the cleaned dataset. Candidate models included Logistic Regression, Random Forest, and Gradient Boosting. The final Power BI dataset includes churn probabilities and risk-level bands so business users can move from analysis to prioritization.





Model	Accuracy	Precision	Recall	ROC-AUC
Logistic Regression	0.692	0.252	0.576	0.715
Random Forest	0.853	0.473	0.181	0.734
Gradient Boosting	0.851	0.419	0.090	0.714

- Best ROC-AUC in recreated evaluation: Random Forest (0.734).
- Logistic Regression provided stronger churn recall and can be useful when the business wants to catch more at-risk customers.
- Tree-based feature importance highlights customer health, inactivity, engagement, usage, and revenue signals as important drivers.

SQL Business Analysis Layer

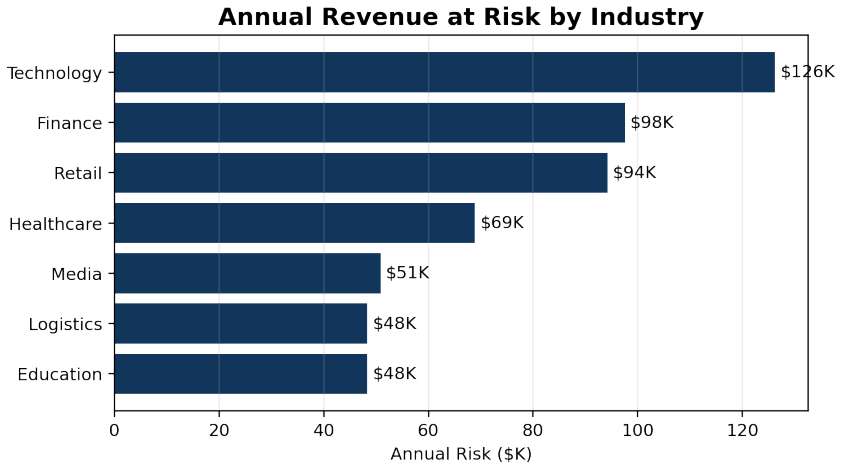
After the Python and modeling phases, the cleaned and enriched dataset was analyzed in SQL. The SQL layer makes the project stronger because it shows how analytics can be converted into repeatable business questions and KPI logic.

SQL Questions Covered

- Executive churn summary: total customers, churned customers, churn rate, annual revenue at risk, and average revenue by customer status.
- Churn by subscription plan, contract type, industry, country, feature adoption, onboarding status, and customer lifetime value segment.
- High-risk customer action list for Customer Success managers based on health score, inactivity, tickets, and revenue value.
- Churn reason analysis with customer profiles such as average revenue, tenure, NPS, tickets, engagement, and inactivity.
- Cohort and time-based analysis by signup year and quarter.

Example KPI Logic

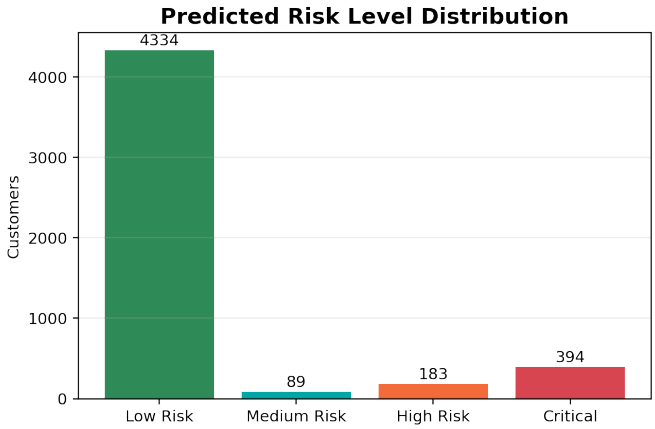
Annual Revenue at Risk = SUM(monthly_charges_usd * 12) for churned customers. This turns churn from a count of lost customers into a business value that executives can understand.



Power BI Dashboard Layer

The Power BI report file contains a four-page executive dashboard. The dashboard is designed to help stakeholders understand churn health, drill into customer segments, identify behavioral risk patterns, and act on predicted high-risk accounts.

Page	Visuals	Primary Business Use
Executive Overview	15	Portfolio-level churn, revenue, KPIs, and filters
Risk Analysis	12	Churn risk by plan, contract, company size, and segments
Behavioral Insights	15	Customer behavior, support burden, engagement, and usage patterns
Predictions	15	ML risk levels, high-risk accounts, and prioritized action lists



Dashboard KPIs to Highlight on GitHu

- Total customers: 5,000
- Observed churn rate: 14.4%
- Total MRR: $429.2K
- Actual annual revenue at risk: $649K
- High/Critical predicted accounts: 577

Recommendations

- Prioritize Customer Success outreach for High Risk and Critical accounts, especially those with high monthly revenue.
- Improve retention strategy for monthly contracts because they show the highest churn rate compared with annual contracts.
- Investigate competitor and missing-feature churn reasons; these are the largest recorded churn drivers.
- Use NPS and customer health score as early-warning signals for proactive retention.
- Monitor industry-level revenue at risk to prioritize strategic segments such as Technology and Finance.

GitHub Report Checklist

- Upload this report to reports/SaaS_Churn_Prediction_Business_Case_Report.pdf.
- Upload the PPTX/PDF deck to presentation/.
- Add dashboard screenshots to screenshots/ so recruiters can preview the dashboard without opening Power BI.
- Add both trained model and scaler artifacts. Recommended names: churn_model.pkl and scaler.pkl.
- In the README, describe the project as a business case study with problem, workflow, insights, recommendations, and links.

Conclusion: This project demonstrates a strong end-to-end analytics workflow: raw data to Python cleaning, feature engineering, model scoring, SQL analysis, and Power BI storytelling. Presented correctly on GitHub, it is a strong portfolio project for Data Analyst, BI Analyst, and junior Analytics Engineering roles.